In-Context Gibbs: Learning from Weak Labels with Frozen Tabular Foundation Models

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Abstract

[TODO: abstract, written last]

1 Introduction

Contributions.

- **Method** (Section 4): In-Context Gibbs, a training-free sampler over latent labels that uses a frozen TFM as its full conditional and handles instance-level (PLL) and group-level (LLP) weak labels with one procedure [C2].
- **Theory** (Section 5): exactness when the TFM is the posterior predictive of its prior [C3]; stability when it is only approximate, even if its conditionals are incompatible [C4]; and a characterisation of when the naive use of a TFM is already sufficient [C12].
- **Evidence from two pre-registered studies** (Section 6): with TabPFN v2 (Study 1) the method did not beat the strongest specialised baselines; a pre-registered replication with TabICL v2 on 28 new datasets (Study 2) did, on both LLP and PLL (Table 1) [C6][C8].
- **Mechanism** (Section 6.5): the advantage comes from modelling dependence between latent labels through the context [C7], and whether it materialises depends on how robust the TFM is to noisy context labels [C13].

[TODO: intro prose]

2 Related work

[TODO: related work prose and Table 1 (positioning)]

3 Problem setup

Data and prior. We observe features $X = (x_1, \dots, x_n)$ whose labels $y = (y_1, \dots, y_n) \in [C]^n$ are latent. As in the prior-fitted networks that underlie tabular foundation models [39], we assume an exchangeable prior: a latent mechanism $\phi \sim p(\phi)$ and $(x_i, y_i) \overset{\text{iid}}{\sim} p(x, y \mid \phi)$, with a test pair (x_*, y_*) drawn from the same mechanism. For a labelled context $D = \{(x_j, y_j)\}_j$, the *posterior predictive distribution* (PPD) of the prior is $p(y_* \mid x_*, D)$; a TFM returns an approximation $q(y_* \mid x_*, D)$ in a single forward pass.

Weak labels. Instead of y we observe $W \sim p(W \mid X, y)$, independent of (x_*, y_*) given (X, y) (assumption A0), through a known observation model $\ell(y) = p(W \mid X, y)$:

- *Partial labels (PLL):* each x_i comes with a candidate set $S_i \ni y_i$, and $\ell(y) = \prod_i \mathbb{1}[y_i \in S_i]$ (the standard assumption that the probability of a candidate set does not depend on which of its members is the true label).
- *Label proportions (LLP):* the points are split into disjoint bags $B_1, \dots, B_G$ and only the label counts w_g of each bag are observed: $\ell(y) = \prod_g \mathbb{1}[\mathrm{counts}(y_{B_g}) = w_g]$.

Positive-unlabelled data, class-dependent label noise and semi-supervised learning are further instance-level cases ($\ell(y) = \prod_i \ell_i(y_i)$); Study 1 evaluated them as secondary settings (Appendix H).

Goal. The target of inference is the posterior over latent labels and the resulting predictive,

$$\pi(y) = p(y \mid X, W) \propto p(y \mid X)\, \ell(y), \qquad p(y_* \mid x_*, X, W) = \sum_y p(y \mid x_*, X, W)\, p(y_* \mid x_*, (X, y)). \tag{1}$$

Neither factor of π is available in closed form for a TFM's prior, and no weight of the TFM may be updated. Our analysis uses three assumptions: (A1) the TFM is exact, $q = p$, on every context the algorithm queries; (A2) the prior's PPD is positive; (A3) ℓ is known and $\sum_y p(y \mid X)\ell(y) > 0$. Theorem 2 removes (A1).

4 In-Context Gibbs

The full conditional is one forward pass. Exchangeability turns a TFM into a Gibbs kernel. For every i and every assignment y with $p(y_{-i} \mid X) > 0$,

$$p(y_i \mid y_{-i}, X) = p(y_i \mid x_i, D_{-i}), \qquad D_{-i} = \{(x_j, y_j)\}_{j \neq i} \tag{2}$$

(Lemma 1): the conditional of one latent label given all the others is the PPD of a query point x_i with the rest of the current assignment as its context. Multiplying by the weak-label factor gives the full conditional of π in (1), $\pi(y_i = k \mid y_{-i}) \propto q(k \mid x_i, D_{-i})\, \ell_i(k)$. A Gibbs sweep over latent labels therefore needs only forward passes of a frozen TFM, and the weak labels enter only through ℓ.

Blocks. One forward pass per label would cost n passes per sweep. We instead partition all n points at random into K blocks in every sweep and, for each block B, query the TFM once with the context D_{-B} (the other blocks, with their current labels) for all points of B:

- *Instance-level weak labels (PLL):* each free y_i, $i \in B$, is drawn from $\propto q(\cdot \mid x_i, D_{-B})\, \ell_i(\cdot)$. Blocks are formed over *all* points, including those whose label is fixed by ℓ: blocking only the free points would bias the composition of the context.

Algorithm 1 In-Context Gibbs (block version; LLP uses bag-blocks). No TFM weight is updated.

Require: features X, weak labels W with model ℓ, frozen TFM q, test points X_*; K blocks, T sweeps, burn-in T_0, S samples

$P_0 \leftarrow$ warm start; draw $y \sim P_0 \cdot \ell$ ▷ LLP: conditioned on bag counts

for $t = 1, \dots, T$ **do**

  partition all points (LLP: all bags) at random into blocks $B_1, \dots, B_K$

  for $k = 1, \dots, K$ **do**

    $Q \leftarrow q(\cdot \mid x_i, D_{-B_k})$ for all $i \in B_k$ ▷ one forward pass

    PLL: draw $y_i \propto Q_i \, \ell_i$ for free $i \in B_k$; LLP: draw $y_{B_g} \sim \prod_{i \in B_g} Q_i$ given counts w_g, for $B_g \subseteq B_k$

    (optional) accept the new y_{B_k} with the MH probability of Proposition 2(a)

  if $t > T_0$ **then** store $y^{(t)}$

return $\hat{p}(y_* \mid x_*) = \frac{1}{S} \sum_s q(y_* \mid x_*, (X, y^{(s)}))$ over S evenly spaced stored samples

- *Label proportions (LLP):* blocks are groups of whole bags (*bag-blocks*). Within each bag, y_{B_g} is drawn from $\prod_{i \in B_g} q(y_i \mid x_i, D_{-B})$ conditioned on the observed counts, exactly, by dynamic programming over counts.

An optional Metropolis–Hastings (MH) correction accepts or rejects the whole block, which makes the block update exact (Proposition 2); it costs one forward pass per changed label. Contexts larger than 2,000 points are subsampled uniformly at random.

Warm start and prediction. The chain starts from a cheap estimate P_0 of the labels: for PLL, the TFM fitted on one candidate per point chosen uniformly at random, cross-fitted over 5 folds; for LLP, linear DLLP [11]. Initial labels are drawn from P_0 restricted by ℓ (for LLP, conditioned on the bag counts). After T sweeps, of which the first T_0 are discarded, the prediction for a test point is the Rao–Blackwellised average over S evenly spaced samples,

$$\hat{p}(y_* \mid x_*) = \frac{1}{S} \sum_{s=1}^{S} q(y_* \mid x_*, (X, y^{(s)})), \tag{3}$$

computed with the TFM's prediction ensemble. This *inductive* predictor is our default. In the *transductive* variant the test points join the chain as free points ($\ell_* \equiv 1$), which is exact in the sense of Theorem 1(c) but enlarges the chain; we evaluate it as an ablation.

Cost and settings. A run costs TK forward passes with a single-estimator TFM plus S passes for the final prediction, and no gradient step. All settings were fixed on development datasets before any evaluation and are the same for every dataset and both studies: $K = 10$, $T = 30$, $T_0 = 10$, $S = 10$, i.e. 310 forward passes per run.

Comparators built on the same TFM. To isolate what the sampler adds, every study also runs, with the same TFM: the *naive TFM*, conditioned on one label per point consistent with the weak labels (a uniformly chosen candidate for PLL; a random count-consistent assignment within each bag for LLP); *hard-EM*, which uses the same warm start and blocks but assigns the constrained arg max instead of sampling (classification EM, 10 iterations, prediction from the final assignment); *stochastic EM*, which predicts from the last sample of the same chain; and an *oracle* conditioned on the true labels, as a ceiling.

5 Theory

We state what holds when the TFM is exact, what survives when it is not, and what the default sampler of Algorithm 1 is *not* covered by. Proofs are in Appendix A.

Lemma 1 (Full conditionals)**.** *Under the exchangeable prior,* (2) *holds for every i and every y with $p(y_{-i} \mid X) > 0$.*

Theorem 1 (Exactness)**.** *Assume (A0)–(A3) and $q = p$. (a) For instance-level weak labels, single-site Gibbs updates $\pi(y_i = k \mid y_{-i}) \propto q(k \mid x_i, D_{-i})\ell_i(k)$ satisfy detailed balance with respect to π, and the chain is irreducible and aperiodic on* $\mathrm{supp}(\pi)$*, hence converges geometrically. (b) For LLP, the MH chain that swaps two differently labelled points of one bag, with acceptance ratio computed from two forward passes, is reversible and irreducible on* $\mathrm{supp}(\pi)$*. (c) The inductive estimator* (3) *converges almost surely to $\bar{p}(y_* \mid x_*) = \sum_y \pi(y)\, p(y_* \mid x_*, (X, y))$, and $|\bar{p} - p(y_* \mid x_*, X, W)| \le \mathrm{TV}\big(p(y \mid X, W),\, p(y \mid x_*, X, W)\big)$, which is zero when the prior is* separable *(the distribution of the features carries no information about $p(y \mid x)$); the transductive chain converges to $p(y_* \mid x_*, X, W)$ exactly.*

Proposition 1 (Bayes optimality under the prior)**.** *For any strictly proper scoring rule, $p(y_* \mid x_*, X, W)$ uniquely minimises the Bayes risk over all measurable predictors of (X, W, x_*), the expectation being taken under the TFM's prior.*

Theorem 1 and Proposition 1 say that, with an exact TFM, In-Context Gibbs computes the Bayes answer for weak labels under the TFM's prior [C3][C10]. This is optimality *on average over the prior*; on real data, which need not follow it, the evidence is empirical (Section 6). A real TFM is only approximately the PPD of any prior, and its conditionals need not even be compatible with a joint distribution. The next result is the one that applies to the sampler as run.

Theorem 2 (Stability under an approximate TFM)**.** *Let P be the kernel of the algorithm with the exact PPD (stationary distribution π) and $\tilde{P}$ the same algorithm with a TFM $q > 0$. Then $\tilde{P}$ has a unique stationary distribution $\tilde{\pi}$, even when the conditionals of q are incompatible, and*

$$\mathrm{TV}(\tilde{\pi}, \pi) \le \tau(P)\, \bar{\varepsilon}, \qquad \tau(P) = \sum_{t \ge 0} \delta(P^t), \quad \bar{\varepsilon} = \sum_y \tilde{\pi}(y)\, \mathrm{TV}\big(\tilde{P}(y, \cdot), P(y, \cdot)\big), \tag{4}$$

where δ is Dobrushin's ergodicity coefficient. For the random-scan single-site kernel, $\bar{\varepsilon} \le \min\{1, 2\kappa\, \varepsilon_{\mathrm{TV}}\}$, where $\varepsilon_{\mathrm{TV}}$ bounds the TV error of the TFM's conditionals and κ depends only on ℓ; for predictions, $|\mathbb{E}_{\tilde{\pi}} q(y_* \mid x_*, \cdot) - \bar{p}(y_* \mid x_*)| \le \varepsilon'_{\mathrm{TV}} + \mathrm{TV}(\tilde{\pi}, \pi)$.

The form of (4) is classical perturbation theory for Markov chains [21, 38, 43, 44], and it is tight in general (equality on a two-state chain). What is specific to our setting is that the chain built from an incompatible TFM still has a unique stationary distribution (it shares the transition graph of the exact chain), and the error terms are expressed through quantities a TFM has, the TV error of its predictive. The bound reads as (integrated mixing time of the exact chain) $\times$ (error per step): a TFM error matters most where the exact chain mixes slowly [C4]. Unlike the analysis of iterative imputation by Liu et al. [35], it is non-asymptotic and covers MH steps.

Proposition 2 (Blocks)**.** *(a) A block update whose proposal is the product of block-wise TFM predictives (for LLP, conditioned on bag counts), accepted with the MH ratio computed by Brook's lemma along the changed labels, leaves π invariant when $q = p$. (b) Without the MH step, Theorem 2 applies with an error $\bar{\varepsilon}_B$ that is bounded only under an extra stability assumption on the PPD, and the bound grows with n for fixed K.*

Scope. The configuration used in our experiments, uncorrected blocks with $K = 10$ and bag-blocks for LLP, is *not* covered by a non-vacuous guarantee: Theorem 1 and Proposition 1 apply to the single-site, swap and MH-corrected samplers, and the bound of Proposition 2(b) is vacuous at our sizes. We therefore report the MH-corrected sampler as an ablation (Section 6.7): at the same number of sweeps it does not predict better than the default, and it is several times slower.

When can the sampler help? The naive use of a TFM conditions it on one label per point drawn from the weak labels. Consider the naive target $\tilde{\eta}_k(x) = P(\tilde{w} = k \mid x)$, the function a consistent learner fitted on these naive labels approaches.

Proposition 3 (When the naive TFM suffices)**.** *(a) If the naive label depends on x only through y, with a channel whose off-diagonal entries are equal within each column, then $\tilde{\eta}_k = b_k + \lambda_k \eta_k$ with $\lambda_k > 0$: the naive target ranks points exactly as the true posterior $\eta_k(x) = p(y = k \mid x)$, and in the binary case, with class prior ϖ, $\mathrm{AUC}(\eta) - \mathrm{AUC}(s) \leq \mathbb{E}|s - \tilde{\eta}|/(\lambda\varpi(1-\varpi))$ for any score s. This covers PU-SCAR, class-dependent noise, missing-completely-at-random SSL, PLL with independent distractors and LLP with random bags. (b) If the naive labels are obtained from the observation by a kernel that does not depend on the parameter, their Fisher information is at most that of the observation, with equality iff the score of the observation is a function of the naive labels. Equality holds for PU-SCAR and label noise; for PLL and LLP the naive labels coarsen the observation and the loss is large (for LLP with bags of 8, about 3% of the information in the bag counts in a logistic model; Appendix A).*

Proposition 3 sorts weak labels into those where the naive TFM already uses all the information in the observation, so that there is nothing for the sampler to recover through ℓ (PU-SCAR, class-dependent noise, SSL), and those where it discards information (PLL, LLP). It predicted, before evaluation, where In-Context Gibbs could outperform the naive TFM; analyses registered as exploratory in Study 1 agreed with this pattern [C12] (Appendix H).

6 Experiments

6.1 Design: two pre-registered studies

Settings and metrics. The main claims concern LLP with random bags of 8 (ROC AUC) and PLL in which each wrong class enters the candidate set independently with probability $q \in \{0.3, 0.6\}$ (accuracy). Weak labels are generated on the training part of stratified 70/30 splits with seeds 0–2; all methods share the splits and the weak labels. Datasets have 500 to 10,000 rows after removing duplicates (Study 2 subsamples one larger dataset to 10,000 by a rule written in advance).

Baselines. For each setting we run its peer-reviewed baselines, each in its published default and tuned by 20 random configurations scored on a validation split *with weak labels only* (the method's own weak-label loss; for LLP the split is by bag). LLP: DLLP [11] with an MLP or a linear model, LLP-VAT [47], EasyLLP [17] and GLWS [18]. PLL: PRODEN [37], GLWS/ILL [18, 20], RC and CC [25]. In-Context Gibbs is not tuned per dataset. The *primary comparator* of each setting was fixed by a rule on development datasets before any evaluation: linear DLLP for LLP and tuned CC for PLL.

Hypotheses and tests. H2 (LLP) and H4 (PLL) require In-Context Gibbs to beat both the primary comparator and the naive TFM (two-sided Wilcoxon signed-rank over datasets, each averaged over 3 seeds, Holm-adjusted), *and* to beat the baseline with the best mean score on at least 70% of datasets (LLP), or to match or beat it on at least 60% of datasets at each level (PLL; Study 1 left the reading of this rule ambiguous, and we report its weakest reading). H3 compares In-Context Gibbs with hard-EM on log loss (Bayesian signed-rank test with a region of practical equivalence of ± 0.005 [13]). A sensitivity analysis repeats every test without datasets flagged in advance for group structure or rule-generated labels.

Two studies. *Study 1* froze the protocol before any evaluation data were used and evaluated 24 OpenML-CC18 datasets (10 binary, 14 multiclass) with TabPFN v2, the TFM registered as primary;

Table 1: Pre-registered confirmatory tests. Δ: mean over datasets of (In-Context Gibbs − comparator), each dataset averaged over 3 seeds; W/T/L: datasets won/tied/lost; p_{Holm}: two-sided Wilcoxon signed-rank, Holm-adjusted within each hypothesis (2 tests for LLP, 4 for PLL). Bold: significant in favour of In-Context Gibbs. "Naive TFM" uses the same TFM as the study. Study 1: 10 binary and 14 multiclass datasets; Study 2: 18 binary and 10 multiclass datasets disjoint from Study 1. H2/H4 additionally required winning against the strongest baseline on $\geq$70%/$\geq$60% of datasets (Section 6.4).

			Study 1: TabPFN v2			Study 2: TabICL v2, new datasets		
Setting	Metric	Gibbs vs.	Δ	W/T/L	p_{Holm}	Δ [95% CI]	W/T/L	p_{Holm}
LLP, bag size 8	AUC	DLLP (linear)	+0.024	8/0/2	0.232	+0.052 [+0.024, +0.086]	16/0/2	$<$**0.001**
		Naive TFM	+0.148	10/0/0	**0.004**	+0.116 [+0.085, +0.150]	18/0/0	$<$**0.001**
PLL, $q = 0.3$	Acc.	CC (tuned)	+0.022	10/0/4	0.084	+0.012 [+0.006, +0.019]	8/2/0	**0.016**
		Naive TFM	+0.099	13/0/1	$<$**0.001**	+0.022 [+0.012, +0.033]	10/0/0	**0.008**
PLL, $q = 0.6$	Acc.	CC (tuned)	−0.132	7/0/7	0.715	+0.012 [+0.006, +0.019]	8/1/1	**0.016**
		Naive TFM	+0.180	12/0/2	**0.002**	+0.205 [+0.094, +0.337]	10/0/0	**0.008**
Pre-registered verdict			H2: not met; H4: not met			H2′: met; H4′: met		

it also registered TabICL v2 as a TFM ablation, secondary weak-label settings, and real partial labels. *Study 2* changed only the TFM. Its hypotheses (H2′, H3′, H4′, identical to Study 1), comparators, tests and dataset-selection rules were fixed in a timestamped commit before any of its evaluation tables was downloaded (Appendix B). Its 28 datasets (18 binary, 10 multiclass) come from curated suites—OpenML100, the AutoML Benchmark, Grinsztajn et al. [27], TabZilla and TabArena—after excluding OpenML-CC18, every Study 1 dataset, synthetic data, and datasets sharing a source with Study 1, and they satisfy the criteria of Study 1. All TFM methods ran on one NVIDIA T4 GPU per run.

6.2 Correctness on problems with an exact posterior

We first checked the theory where the exact posterior can be enumerated ($n \leq 14$) and the PPD is analytic. With the exact PPD as q, the stationary distributions of the single-site, swap and MH-corrected block kernels match π to within 3×10^{-14} in total variation for every weak-label type and for a prior in which features do or do not carry information about the labels; running the actual sampler for 10^5 sweeps reaches the Monte Carlo floor. With small PFNs trained on the same prior, the stationary distribution is within 0.012 (single-site) of π, and the bound of Theorem 2 held in all 42 configurations, including 30 with deliberately incompatible conditionals, while being 2–12 times loose. Blocking over all points with $K = 10$ changes predictions by 1.7×10^{-4} at $n = 1{,}000$ on an exactly solvable pooled problem, but blocks of two points on tiny problems with informative features move the posterior by up to 0.04 (Appendix E). [TODO: RQ1 numbers are typed from G4/RQ1_REPORT; generate them by script (results/rq1/*.csv) before submission]

6.3 Study 1: TabPFN v2

Study 1 evaluated the frozen protocol with TabPFN v2 [29] as the conditional (Table 1, left). In-Context Gibbs improved on the naive use of the same TFM in both main settings, for example by +0.148 AUC on LLP, but it did not beat the strongest specialised baselines. On LLP with bags of 8 it was +0.024 AUC ahead of linear DLLP (8/0/2, $p_{\text{Holm}} = 0.232$). On PLL it was ahead of tuned CC at $q = 0.3$ (+0.022, $p_{\text{Holm}} = 0.084$) and far behind at $q = 0.6$ (-0.132), where the chains collapsed on the four datasets with at least eight classes (Figure 2, bottom). Both pre-registered main hypotheses therefore failed; the mechanism hypothesis H3 held (Table 2).

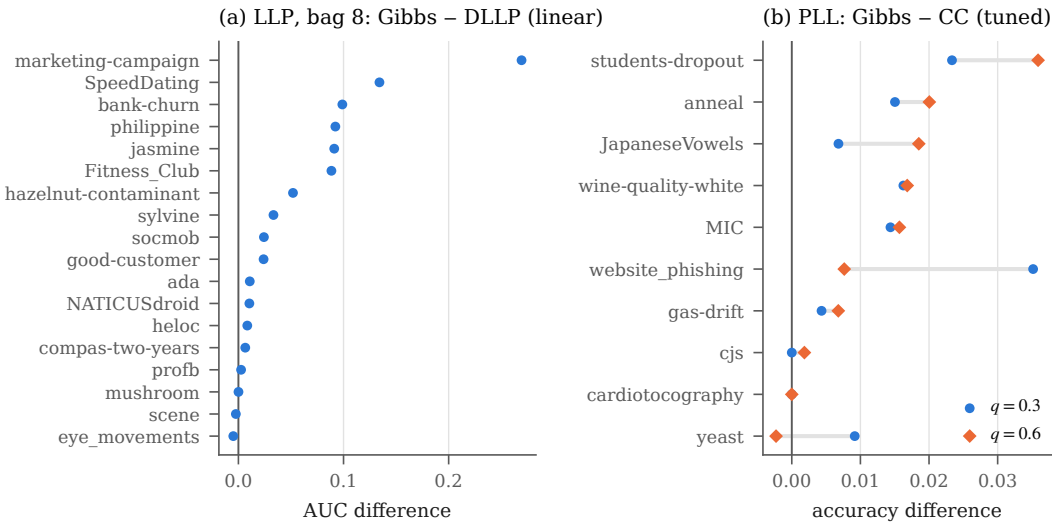


Figure 1: Study 2, per dataset: difference between In-Context Gibbs and the primary comparator (positive favours In-Context Gibbs), mean over 3 seeds. (a) LLP with bags of 8, AUC against linear DLLP. (b) PLL, accuracy against tuned CC at two candidate-set rates; a horizontal bar joins the two rates of a dataset.

6.4 Study 2: a pre-registered replication with TabICL v2 on new datasets

In an ablation of Study 1, replacing TabPFN v2 by TabICL v2 [8] passed every test (Appendix C). Because that TFM was chosen after seeing the results, we treated the finding as a hypothesis and tested it in Study 2, changing nothing but the TFM, on 28 datasets disjoint from those of Study 1. All pre-registered tests passed (Table 1, right; Figure 1). On LLP with bags of 8, In-Context Gibbs improved AUC over linear DLLP by +0.052 [+0.024, +0.086] (16/0/2, $p_{\text{Holm}} < 0.001$) and beat the strongest LLP baseline, tuned DLLP (MLP), on 16/18 datasets (required: $\geq$70%) [C6]. On PLL it improved accuracy over tuned CC by +0.012 at $q = 0.3$ (8/2/0) and by +0.012 at $q = 0.6$ (8/1/1; $p_{\text{Holm}} = 0.016$ for both), and was at least as accurate as the strongest PLL baseline on 10/10 and 9/10 datasets (required: $\geq$60% at each level) [C8]. Unlike in Study 1, it did not collapse on the four datasets with at least eight classes. The PLL gains are modest in absolute terms: In-Context Gibbs is within 0.8 and 1.9 accuracy points of an oracle trained on the true labels, so the room for improvement is small. Removing datasets flagged for group structure or temporal drift did not change any conclusion (Appendix D).

6.5 Why it works, and when it does not

Sampling versus hard assignment. With the same TFM and the same warm start, In-Context Gibbs had lower log loss than hard-EM in 86.7% of 128 cells in Study 1 and 78.9% of 38 cells in Study 2, and the advantage grew with the weakness of the labels (Table 2) [C7].

Robustness of the TFM to noisy context labels. Figure 2 (top) measures what one Gibbs sweep does to an assignment in which a fraction r of the training labels is correct (PLL, $q = 0.6$; exploratory, Study 1 datasets only). With TabPFN v2 the gain $\Phi(r) - r$ is negative over a range of r on all four datasets with at least eight classes, so a chain that starts near the uninformative state is pushed back towards it; with TabICL v2 the gain is positive everywhere. The chains behave accordingly (bottom): TabPFN chains stay near chance on mfeat-factors and elsewhere escape slowly, or not at all, depending on the seed, whereas TabICL chains converge within about ten sweeps [C13]. [TODO: link to the collapse-threshold analysis (G9_PLAN D2), if a proposition is ready by 20/12]

Table 2: In-Context Gibbs versus hard-EM with the same TFM and warm start, log loss. A cell is a (dataset, weak-label level) pair; Bayesian signed-rank test with ROPE ± 0.005; Spearman correlation between weakness level and the gap. †Computed on PLL cells (LLP has a single level), as stated before the analysis was run.

Study (TFM; settings)	cells	Gibbs better	P(Gibbs better)	P(ROPE)	Spearman ρ
Study 1 (TabPFN v2; all six settings)	128	86.7%	1.00	0.00	0.36
Study 2 (TabICL v2; LLP bag 8, PLL)	38	78.9%	1.00	0.00	0.66†

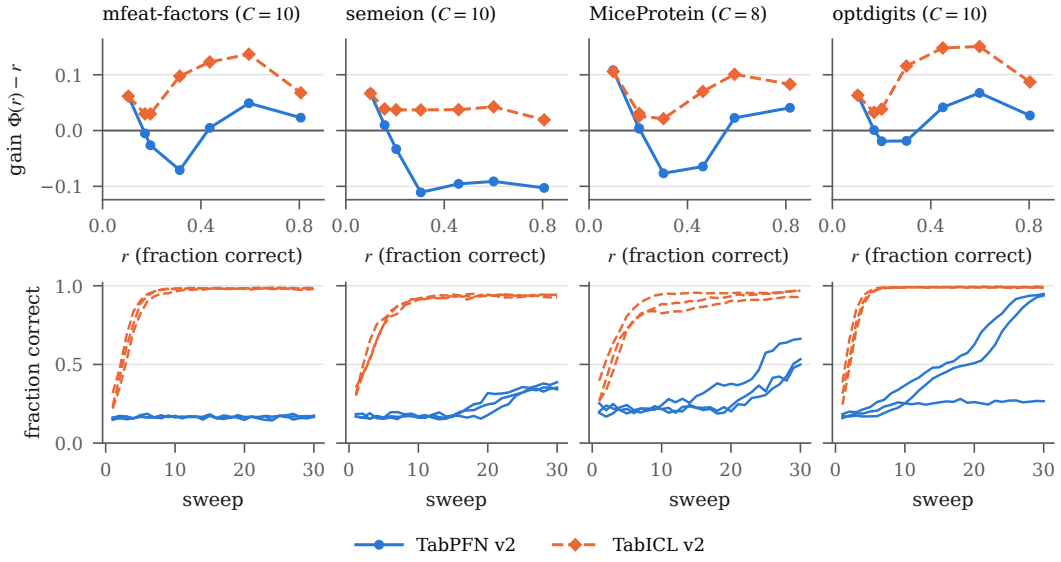


Figure 2: PLL at $q = 0.6$ on the four Study 1 datasets with at least eight classes. Top: gain $\Phi(r) - r$ of one Gibbs sweep started from an assignment with a fraction r of correct labels (mean over 3 seeds; exploratory). Bottom: fraction of correct training labels along the chains of the main runs, one line per seed (diagnostic only; the sampler never sees the true labels).

6.6 Real weak labels

On three real partial-label datasets (BirdSong, Lost, MSRCv2; 10 random 90/10 splits; TabICL v2, as registered in Study 1 because these datasets have more than ten classes), In-Context Gibbs had the best mean accuracy on all three and the best mean rank among 10 methods (2.15; Table 3). Its margin over the best baseline was +0.028 on BirdSong (8/2 splits won/lost) and +0.045 on MSRCv2 (8/2), but only +0.002 on Lost (5/5), where it ranked 3rd by mean rank. Real candidate sets are small (2.2–3.2 labels on average).

6.7 Ablations, mixing and cost

Exact versus approximate sampler. The MH-corrected block sampler, which is exact when the TFM is (Proposition 2), did not give systematically better predictions than the default uncorrected sampler at the same 30 sweeps (Table 5). With TabPFN v2 (Study 1) the differences were +0.010 AUC on LLP (4/0/5 cells) and +0.006 accuracy on PLL at $q = 0.3$ (4/3/2). With TabICL v2 (exploratory, Study 1 datasets) they were +0.033 AUC on LLP (6/0/3; more than half of it from one dataset on which the default chain was close to chance, and log loss worse by 0.064), -0.009 and -0.021 accuracy on PLL at $q = 0.3$ and 0.6, and -0.016 on datasets with at least eight classes at $q = 0.6$ (0/0/5). The corrected sampler accepted as few as 0.12 of its block proposals and was a median 7

Table 3: Real partial labels (Study 1, TabICL v2 for all TFM-based methods): accuracy averaged over 10 random 90/10 splits, and mean rank over all splits of the three datasets among the ten ranked methods (In-Context Gibbs, naive TFM and eight baselines). Bold: best non-oracle accuracy.

Method	BirdSong	Lost	MSRCv2	mean rank
In-Context Gibbs (ours)	**0.759**	**0.750**	**0.628**	2.15
Hard-EM (same TFM)	0.730	0.665	0.627	–
Stochastic EM (same TFM)	0.744	0.716	0.622	–
Naive TFM	0.689	0.481	0.491	7.27
PRODEN (default)	0.667	0.373	0.344	9.88
PRODEN (tuned)	0.707	0.748	0.465	5.12
GLWS (default)	0.722	0.712	0.582	3.98
GLWS (tuned)	0.731	0.747	0.490	4.03
RC (default)	0.719	0.720	0.424	6.00
RC (tuned)	0.702	0.721	0.478	5.23
CC (default)	0.719	0.717	0.425	6.02
CC (tuned)	0.700	0.723	0.479	5.32
Oracle (true labels)	0.835	0.889	0.824	–

times slower (up to 206 times). At a fixed budget it mixes more slowly, so this comparison cannot separate the bias of the approximation from the slower mixing of the exact chain; it does show that the cheaper sampler loses nothing in prediction. Moreover, exactness holds only under (A1): with a real TFM, whose conditionals are incompatible, the corrected chain does not target a well-defined posterior either. In the same vein, the transductive predictor, which converges to the exact Bayes predictive (Theorem 1(c)), was *worse* than the inductive one on LLP (-0.013 AUC, 7/0/23; log loss worse by 0.042).

Other components. Changing the number of blocks ($K = 5$ or 20), the number of sweeps (15 or 60) or the initialisation (cold start from ℓ) changed AUC or accuracy by at most about 0.015 on average (Appendix F).

Mixing. With two chains per cell, the median fraction of training labels with $\widehat{R} > 1.1$ [48] was 11% (TabPFN) and 28% (TabICL) on LLP, and 21% and 31% on PLL at $q = 0.6$, against at most about 4.5% for perfectly mixed chains; yet the median difference between the test metrics of the two chains was at most 0.041 (Table 5), and 0.002 on the four datasets with at least eight classes (TabICL). Thirty sweeps give stable predictions, not samples from a converged posterior over individual labels.

Cost. No weight is trained. On one T4 GPU, a run of In-Context Gibbs took a median of 41 s (maximum 5.3 min) with TabICL v2 in Study 2, for up to 7,000 training points, and a median of 197 s (maximum 27.5 min) with TabPFN v2 in Study 1; peak GPU memory was 3.7 GiB and 1.9 GiB [C11].

7 Limitations

The TFM decides. The same sampler did not beat the strongest specialised baselines with TabPFN v2 (Study 1) and did with TabICL v2 (Study 2). The two TFMs were compared on the same data only in Study 1, where TabICL v2 was an ablation, and in the exploratory response-map analysis; the studies use different datasets, so they do not form a controlled comparison of TFMs. We do not claim the method works with any TFM; practitioners should check the response of their TFM to noisy context labels (Figure 2) before relying on it.

Theory and the default sampler. Our guarantees cover the single-site, swap and MH-corrected samplers. The uncorrected block and bag-block samplers used by default have no non-vacuous guarantee (Proposition 2); the exact variant does not predict better at the same budget, but it also mixes more slowly, so the bias of the approximation itself has not been measured on real data. Exactness also holds only with respect to the TFM's prior, and the exact transductive predictor was worse than the inductive one.

Mixing. Thirty sweeps give predictions that are stable across chains, but label-level $\widehat{R}$ shows that the chains have not mixed over individual labels, so the samples should not be read as draws from a converged posterior.

Scope and size of the gains. The claims cover LLP with bags of 8 and PLL. In Study 1, In-Context Gibbs did not improve on the naive TFM for PU with labels selected completely at random, class-dependent noise or semi-supervised learning, as Proposition 3 predicts, and it was not significantly better than the primary comparator for bags of 16 or 32. On PLL the gain over tuned CC is about one accuracy point, close to an oracle ceiling. Datasets have at most 10,000 rows and contexts at most 2,000 points; one baseline (default PRODEN) failed deterministically on one Study 2 dataset.

8 Conclusion

[TODO: conclusion: only claims already supported in the paper]

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A Proofs

Assumptions. (A0) $W \sim p(W \mid X, y)$, independent of (x_*, y_*) given (X, y); nuisance parameters of the observation model, if any, are a priori independent of (ϕ, X, y). (A1) $q(\cdot \mid x, D) = p(\cdot \mid x, D)$ for every context the algorithm queries. (A2) $p(k \mid x, D) > 0$ for all k and every context of positive probability. (A3) ℓ is known and $\sum_y p(y \mid X)\ell(y) > 0$. For instance-level observations $\ell(y) = \prod_i \ell_i(y_i)$; the set of free points is $F = \{i : |\{k : \ell_i(k) > 0\}| \geq 2\}$.

A.1 Lemma 1

Proof of Lemma 1. Given ϕ the pairs (x_j, y_j) are i.i.d., hence exchangeable, so the joint law of $(D_{-i}, (x_i, y_i))$ equals that of $(D_{n-1}, (x_*, y_*))$, a context of $n-1$ points and a query. The information (X, y_{-i}) coincides with (x_i, D_{-i}). Therefore $p(y_i \mid y_{-i}, X) = p((x_i, y_i), D_{-i}) / \sum_k p((x_i, k), D_{-i}) = p(y_i \mid x_i, D_{-i})$. □

The PPD conditions on the query features x_* as well, which may carry information about ϕ; this is the quantity the PFN training loss targets, since x_* is drawn from the prior together with the context [39].

Corollary 1 (Positivity)**.** *Under (A2),* $p(y \mid X) > 0$ *for every* $y \in [C]^n$.

Proof. Some y^0 has $p(y^0 \mid X) > 0$. Change one coordinate at a time, $y^0 \to y^1 \to \cdots \to y^m = y$, step k changing point s_k. If $p(y^{k-1} \mid X) > 0$, then $p(y^{k-1}_{-s_k} \mid X) > 0$, the context $D^{k-1}_{-s_k}$ has positive probability, and by Lemma 1 and (A2) $p(y^k \mid X) = p(y^{k-1}_{-s_k} \mid X)\, p(y^k_{s_k} \mid x_{s_k}, D^{k-1}_{-s_k}) > 0$. Induct on k. □

A.2 Theorem 1

Proof of Theorem 1(a). The update of point i is $P_i(y, y') = \pi_i(y'_i \mid y_{-i}) \mathbb{1}[y'_{-i} = y_{-i}]$ with $\pi_i(k \mid y_{-i}) \propto q(k \mid x_i, D_{-i})\ell_i(k)$. By Lemma 1 and (A1), $\pi_i(k \mid y_{-i}) \propto p(k \mid y_{-i}, X)\ell_i(k) \propto \pi(k, y_{-i})$, the full conditional of π; hence $\pi(y)P_i(y, y') = \pi(y_{-i})\pi_i(y_i \mid y_{-i})\pi_i(y'_i \mid y_{-i})$ is symmetric in (y, y'). By Corollary 1, $\pi(y) > 0$ iff $\ell(y) > 0$, a product set; any two of its points are joined by changing one coordinate at a time inside the set, each step with probability at least $\pi_i(\cdot \mid \cdot)/|F| > 0$, so the random-scan kernel $P = |F|^{-1} \sum_{i \in F} P_i$ is irreducible, and aperiodic since $P(y, y) > 0$. A finite, irreducible, aperiodic chain converges geometrically. The implementation sweeps the free points in a uniformly random order, $P_{\text{sw}} = \frac{1}{|F|!} \sum_\sigma P_{\sigma(1)} \cdots P_{\sigma(|F|)}$; each factor preserves π, and

P_{sw} is reversible because the adjoint of a product is the product in reverse order and the set of permutations is closed under reversal. □

Proof of Theorem 1(b). Choose a bag uniformly; if it is not pure, choose uniformly an ordered pair (i, j) in it with $y_i \neq y_j$ and propose y' by swapping their labels. Brook's lemma through the intermediate state (y'_i, y_{-i}) gives

$$\frac{p(y' \mid X)}{p(y \mid X)} = \frac{p(y'_i \mid y_{-i}, X)}{p(y_i \mid y_{-i}, X)} \cdot \frac{p(y'_j \mid y'_i, y_{-ij}, X)}{p(y_j \mid y'_i, y_{-ij}, X)},$$

and Lemma 1 with (A1) turns the right-hand side into the ratio R_{ij} of four TFM predictives that the algorithm computes with two forward passes. A swap preserves the counts, so $\ell(y') = \ell(y)$ and $R_{ij} = \pi(y')/\pi(y)$, independently of the order of (i, j). The number of differently labelled pairs depends only on the counts, so the proposal is symmetric and the chain is Metropolis with stationary distribution π. Two assignments of a bag with the same counts are joined by a sequence of swaps preserving the counts, each with positive probability by Corollary 1; bags are independent, so the product chain is irreducible (the lazy version is also aperiodic). □

Proof of Theorem 1(c). (Inductive.) The ergodic theorem for a finite irreducible chain applied to the bounded function $y \mapsto p(y_* \mid x_*, (X, y))$, with (A1), gives almost sure convergence of (3) to $\bar{p}$. By the law of total probability and (A0), $p(y_* \mid x_*, X, W) = \sum_y p(y \mid x_*, X, W)\, p(y_* \mid x_*, (X, y))$; the difference of the expectations of one $[0, 1]$-valued function under two measures is at most their TV distance. If the prior is separable, $\phi = (\phi_x, \phi_{y|x})$ independent a priori with $p(x \mid \phi) = p(x \mid \phi_x)$ and $p(y \mid x, \phi) = p(y \mid x, \phi_{y|x})$, then $p(y \mid x_*, X, W) \propto \ell(y) \int p(y \mid X, \phi_{y|x}) p(\phi_{y|x}) d\phi_{y|x}$ times a factor that does not depend on y, so $p(y \mid x_*, X, W) = p(y \mid X, W)$ and the bound is zero. (Transductive.) Adding x_* as a free point with $\ell_* \equiv 1$ gives a chain for $p(y, y_* \mid X, x_*, W)$ that is exact by part (a) (for LLP, the swap chain combined with single-site updates of x_*, each preserving the target). By Lemma 1, the Rao–Blackwell term is $q(y_* \mid x_*, (X, y^{(s)}))$, whose average converges to $p(y_* \mid x_*, X, W)$. □

Proof of Proposition 1. By the tower property, $\mathbb{E}[S(f, y_*)] = \mathbb{E}\big[\mathbb{E}[S(f, y_*) \mid X, W, x_*]\big]$. For each value of (X, W, x_*), f is a fixed distribution and $y_* \sim p(\cdot \mid x_*, X, W)$; strict propriety gives the unique minimiser $f = p(\cdot \mid x_*, X, W)$. Its risk is finite (at most $\log C$ for the log score). □

A.3 Theorem 2

Proof of Theorem 2. Existence and uniqueness. The state space is finite, so $\tilde{P}$ has a stationary distribution. With $q > 0$ every single-site update and every MH acceptance probability is positive, so $\tilde{P}$ has the transition graph of P and is irreducible; $\tilde{\pi}$ is unique. Compatibility of the conditionals of q is never used. *Perturbation identity.* Let $\Delta = \tilde{\pi} - \pi$ and $\nu = \tilde{\pi}(\tilde{P} - P)$, so $\nu(\Omega) = 0$. From $\tilde{\pi} = \tilde{\pi}\tilde{P}$ and $\pi = \pi P$, $\Delta = \nu + \Delta P$, and iterating, $\Delta = \sum_{t<T} \nu P^t + \Delta P^T$. *Contraction.* For a signed measure μ with $\mu(\Omega) = 0$, $\|\mu K\|_1 \leq \delta(K)\|\mu\|_1$. If $\tau(P) < \infty$ then $\delta(P^T) \to 0$, the remainder vanishes, and $\|\Delta\|_1 \leq \|\nu\|_1 \tau(P)$; if $\tau(P) = \infty$ there is nothing to prove (no aperiodicity is needed). Finally $\|\nu\|_1 \leq \sum_y \tilde{\pi}(y) \|\tilde{P}(y, \cdot) - P(y, \cdot)\|_1 = 2\bar{\varepsilon}$. *Single-site error.* $\tilde{P}(y, \cdot) - P(y, \cdot)$ averages the differences of the single-site updates, and each is the TV distance between $q\ell_i$ and $p\ell_i$ after normalisation. With $Z_p = \sum_k p(k)\ell_i(k)$, $2\,\mathrm{TV}(q_\ell, p_\ell) \leq \big(\sum_k \ell_i(k)|q(k) - p(k)| + |Z_q - Z_p|\big)/Z_p \leq 4(\max_k \ell_i(k)/Z_p)\,\mathrm{TV}(q, p)$, so the error is at most $2\kappa\varepsilon_{\mathrm{TV}}$ with $\kappa = \sup \max_k \ell_i(k)/Z_p$. *Prediction.* $|\mathbb{E}_{\tilde{\pi}} q - \mathbb{E}_\pi p| \leq |\mathbb{E}_{\tilde{\pi}}(q - p)| + |\mathbb{E}_{\tilde{\pi}} p - \mathbb{E}_\pi p| \leq \varepsilon'_{\mathrm{TV}} + \mathrm{TV}(\tilde{\pi}, \pi)$. □

Relation to classical bounds. With $E = \tilde{P} - P$ and the fundamental matrix Z, $\tilde{\pi} - \pi = \nu Z$ [38, 43], which gives $\mathrm{TV}(\tilde{\pi}, \pi) \leq \tau_1(Z)\bar{\varepsilon}$ with $\tau_1(Z) \leq \tau(P)$ [21, 44]; (4) is the weaker, more interpretable form. It is attained by a two-state chain. For a single-site sweep the per-sweep error is at most $|F|$ times the per-update error; the MH swap kernel has error at most $4\varepsilon_\infty$, with $\varepsilon_\infty = \sup|\log q - \log p|$.

A.4 Proposition 2 (blocks)

Proof of Proposition 2. (a) The proposal Q_B depends only on y_{-B}, which the step does not change, so the update is MH-within-Gibbs with an independent proposal. Brook's lemma along the changed labels with Lemma 1 gives the exact ratio $p(y' \mid X)/p(y \mid X)$, multiplied by the ratio of ℓ (one for LLP). The proposal is positive on the support of the block and acceptance is positive, so the chain is irreducible. (b) Let P_{bG} be exact block Gibbs. Under the stability assumption that adding one labelled point or one unlabelled feature to a context changes the PPD by at most β in TV, the chain rule and a hybrid argument give $\mathrm{TV}\big(\prod_{i\in B\cap F} p(\cdot \mid x_i, D_{-B}),\, p(y_{B\cap F} \mid y_{-(B\cap F)}, X)\big) \le m_f(m-1)\beta$ for a block of m points of which m_f are free, and after reweighting by ℓ, $\bar{\varepsilon}_B \le 2\kappa_B\, m_f(m-1)\beta$; Theorem 2 with $P = P_{\mathrm{bG}}$ concludes. □

Why this is vacuous at our sizes. With $\beta = O(1/n)$ and fixed K ($m = n/K$), $m_f(m-1)\beta = O(n/K^2)$ grows with n. For bag-blocks, κ_B is the inverse probability that all bags of a group match their counts under the product proposal: for $n \approx 2{,}000$ and $K = 10$, $\log_{10}\kappa_B \approx 15.8$ for bags of 8. On an exactly solvable LLP problem (3 bags of 4), the uncorrected bag-block chain has $\mathrm{TV}(\tilde{\pi}, \pi)$ between 0 and about 0.6 depending on the prior and grouping, so the approximation is real at small n; at our sizes we measure its effect on predictions instead (Section 6.7).

A.5 Proposition 3 (when the naive TFM suffices)

Proof of Proposition 3. (a) With $P(\tilde{w} = k \mid x, y) = M_{y,k}$ independent of x and $M_{y,k} = b_k$ for $y \neq k$, $\tilde{\eta}_k = M_{k,k}\eta_k + b_k(1-\eta_k) = b_k + \lambda_k\eta_k$, $\lambda_k = M_{k,k} - b_k$. For PLL with each wrong class entering the candidate set independently with probability q and the naive label chosen uniformly from the set, $M_{y,y} = a = \mathbb{E}[1/(1+N)] = \frac{1-(1-q)^C}{Cq}$ with $N \sim \mathrm{Bin}(C-1, q)$ and $b = (1-a)/(C-1)$, so $\lambda = (Ca-1)/(C-1) > 0$ for $q < 1$. For LLP with random bags of size k, $P(\tilde{w}_i = 1 \mid x_i, y_i) = (y_i + (k-1)\varpi)/k$, so $\lambda = 1/k$. A strictly increasing affine map preserves ranks, hence one-vs-rest AUC. For the binary bound, write $\varpi(1-\varpi)\mathrm{AUC}(s) = \mathbb{E}[\eta(X)(1-\eta(X'))h_s(X,X')]$ for independent copies, with $h_s = \mathbb{1}[s(X) > s(X')] + \frac{1}{2}\mathbb{1}[s(X) = s(X')]$; symmetrising, $\varpi(1-\varpi)[\mathrm{AUC}(\eta) - \mathrm{AUC}(s)] = \frac{1}{2}\mathbb{E}\big[|\Delta|(\mathbb{1}[s \text{ reverses the pair}] + \frac{1}{2}\mathbb{1}[s \text{ ties}, \Delta \neq 0])\big]$ with $\Delta = \eta(X) - \eta(X')$. With $t = (s-b)/\lambda$, which has the ranks of s, a reversed or tied pair has $|\Delta| \le |\eta(X) - t(X)| + |\eta(X') - t(X')|$, so the right-hand side is at most $\mathbb{E}|s - \tilde{\eta}|/\lambda$. For $\lambda = 1$ this is a known AUC regret bound [1, Corollary 12]. (b) $p_\theta(\tilde{w}) = \sum_O p_\theta(O)R(\tilde{w} \mid O)$ gives $\partial_\theta \log p_\theta(\tilde{w}) = \mathbb{E}_\theta[\partial_\theta \log p_\theta(O) \mid \tilde{w}]$, and Jensen's inequality for conditional expectations gives $I_{\tilde{w}}(\theta) \le I_O(\theta)$, with equality iff the score of O is $\sigma(\tilde{w})$-measurable. □

Information lost by the naive labels (logistic model $\eta = \sigma(\theta x)$, $\theta = 2$, $x \sim \mathcal{N}(0, I)$; per point). LLP: the Godambe information of the pointwise (composite) likelihood on naive labels, relative to the information of the bag counts, is 0.134, 0.060, 0.033, 0.017, 0.009 for bags of 2, 4, 8, 16, 32. PLL: the naive labels retain 0.24–0.26 of the information in the candidate sets at $q = 0.3$ and 0.06–0.10 at $q = 0.6$ ($C \in \{3, 4, 7\}$). For PU-SCAR and label noise the ratio is 1.

B Pre-registration and timeline

[TODO: timeline table]

C Study 1 with TabICL v2 (exploratory)

[TODO: G8_REPORT §2, §6.4]

D Study 2: sensitivity analysis and full results

Table 4: Study 2: mean score of every method (datasets averaged over 3 seeds). PRODEN (default) failed on gas-drift in 5 of 6 cells (non-finite probabilities, reproduced on a rerun); its $q = 0.3$ mean is over 9 datasets and its $q = 0.6$ gas-drift entry uses one seed.

LLP, bag 8 (AUC, 18 datasets)			PLL (accuracy, 10 datasets)				
Method	default	tuned	Method	$q = 0.3$		$q = 0.6$	
				default	tuned	default	tuned
In-Context Gibbs (ours)	0.840		**In-Context Gibbs (ours)**	0.862		0.851	
Hard-EM (same TFM)	0.827		Hard-EM (same TFM)	0.862		0.803	
Stochastic EM (same TFM)	0.835		Stochastic EM (same TFM)	0.858		0.843	
Naive TFM	0.724		Naive TFM	0.840		0.647	
DLLP (MLP)	0.783	0.799	PRODEN	0.800	0.836	0.678	0.820
DLLP (linear)	0.788	0.794	GLWS	0.814	0.848	0.724	0.837
LLP-VAT	0.778	0.798	RC	0.828	0.849	0.803	0.839
EasyLLP	0.724	0.771	CC	0.827	0.850	0.804	0.839
GLWS	0.784	0.789	Oracle (true labels)	0.870		0.870	
Oracle (true labels)	0.879						

[TODO: S1 sensitivity (S2_RESULTS §1, §2), per-baseline comparisons (§4), per-dataset tables]

E Correctness experiments (RQ1)

[TODO: E1–E5 tables and figure from G4/RQ1_REPORT (generated from results/rq1/*.csv)]

F Ablations

Table 5: Mixing and the exact sampler, by TFM. TabPFN: Study 1 (registered ablations); TabICL: exploratory lot on the Study 1 datasets. "3 small": the three smallest datasets of each group, as registered for $\widehat{R}$. Columns 2–5: medians over cells of the fraction of training labels with $\widehat{R} > 1.1$ (two chains) and of the absolute difference between the test metrics of the two chains. Columns 6–7: mean difference in AUC (LLP) or accuracy (PLL) between the MH-corrected and the default sampler at 30 sweeps, with the mean MH acceptance rate. In the last row, $\widehat{R}$ uses all four datasets with $\geq$8 classes, and the MH comparison only two of them (mfeat-factors, optdigits; 5 of 12 cells), because the corrected sampler exceeded the 12-hour compute session on optdigits.

	labels with $\widehat{R} > 1.1$		test-metric gap, 2 chains		MH-corrected − default	
Setting (datasets)	TabPFN	TabICL	TabPFN	TabICL	TabPFN	TabICL [acc. rate]
LLP, bag 8 (3 small)	11%	28%	0.041	0.018	+0.010	+0.033 [0.68]
PLL, $q = 0.3$ (3 small)	6%	6%	0.006	0.018	+0.006	−0.009 [0.78]
PLL, $q = 0.6$ (3 small)	21%	31%	0.016	0.021	–	−0.021 [0.72]
PLL, $q = 0.6$, $\geq$8 classes (4; MH: 2)	–	22%	–	0.002	–	−0.016 [0.47]

[TODO: full ablation table (G8/results/ablation_summary.csv) and size curve]

G Datasets, baselines and tuning spaces

[TODO: dataset cards; docs/tuning_spaces.md]

H Secondary settings of Study 1

[TODO: LLP bags 16/32, PU-SCAR, PU-SAR, noisy labels, SSL; D1-D3]